

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1558
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COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I M Vishnu Vardhan Reddy (192572062)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Vishnu

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

Demand Forecasting

- ARIMA
- Prophet
- LSTM Neural Networks

Recommendation System

- Collaborative Filtering
- Content-Based Filtering
- Hybrid Recommendation

Customer Segmentation

- K-Means Clustering
- RFM Analysis

Benefits

- Prevents stock shortages
- Reduces overstock
- Improves customer satisfaction
- Increases sales

Task 2: Social Media Fraud Detection Problem

Problem Statement

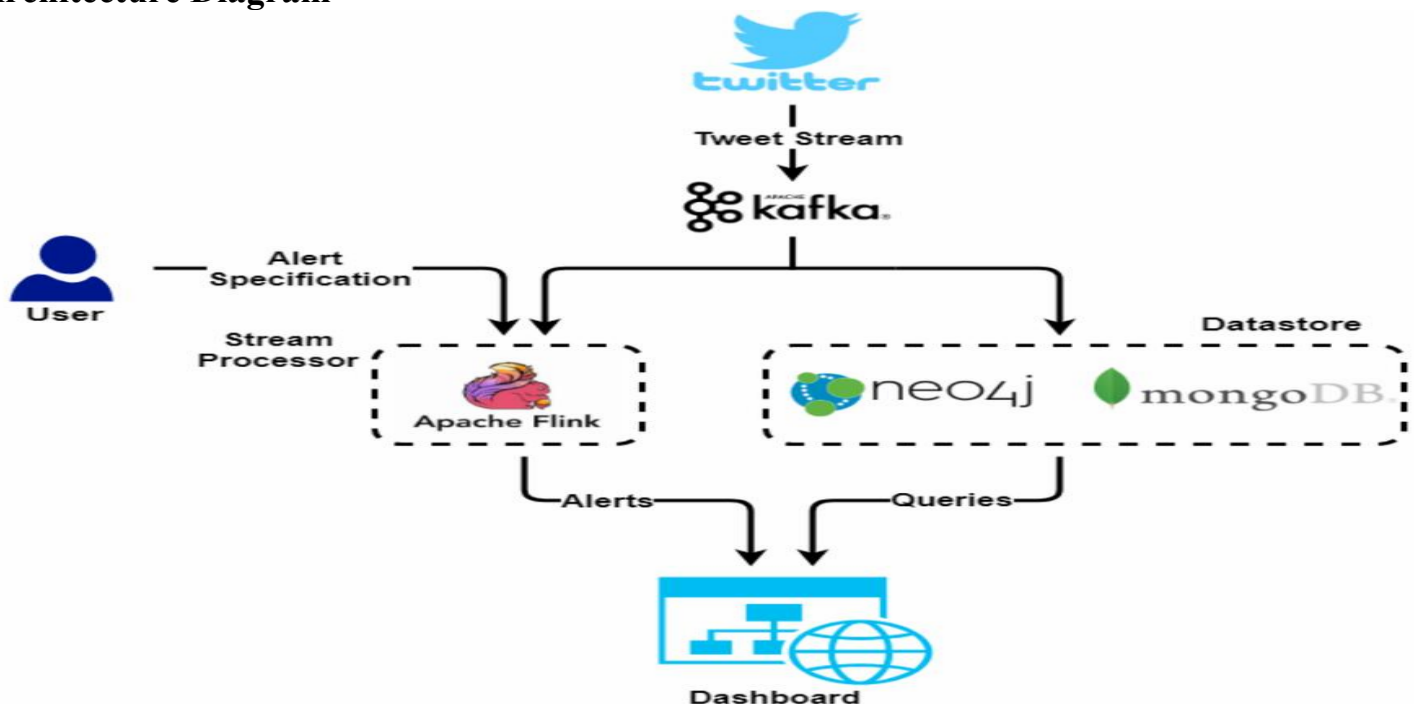
A social media platform processes billions of posts, comments, likes, and shares every day. It needs to detect:

- Fake accounts
- Spam
- Misinformation

Objective

Build a scalable big data system capable of real-time fraud detection.

Architecture Diagram



Architecture Components

Component	Technology
Data Sources	Posts, Comments, Likes

Message Queue	Apache Kafka
Stream Processing	Apache Flink
Storage	HDFS
ML Models	TensorFlow
Visualization	Grafana

Workflow

1. User activities are streamed into Kafka.
2. Flink processes events instantly.
3. Features are extracted.
4. ML models classify suspicious behavior.
5. Alerts are generated.

Machine Learning Models

Fake Account Detection

- Random Forest
- XGBoost

Spam Detection

- Naïve Bayes
- Logistic Regression

Misinformation Detection

- BERT Transformer
- NLP Models

Stream Processing Role

Apache Flink provides:

- Low latency
- Continuous processing
- High throughput

Benefits

- Detects fraud instantly
- Reduces fake accounts
- Improves platform trust

Task 3: Government National Data Platform

Problem Statement

A government agency combines:

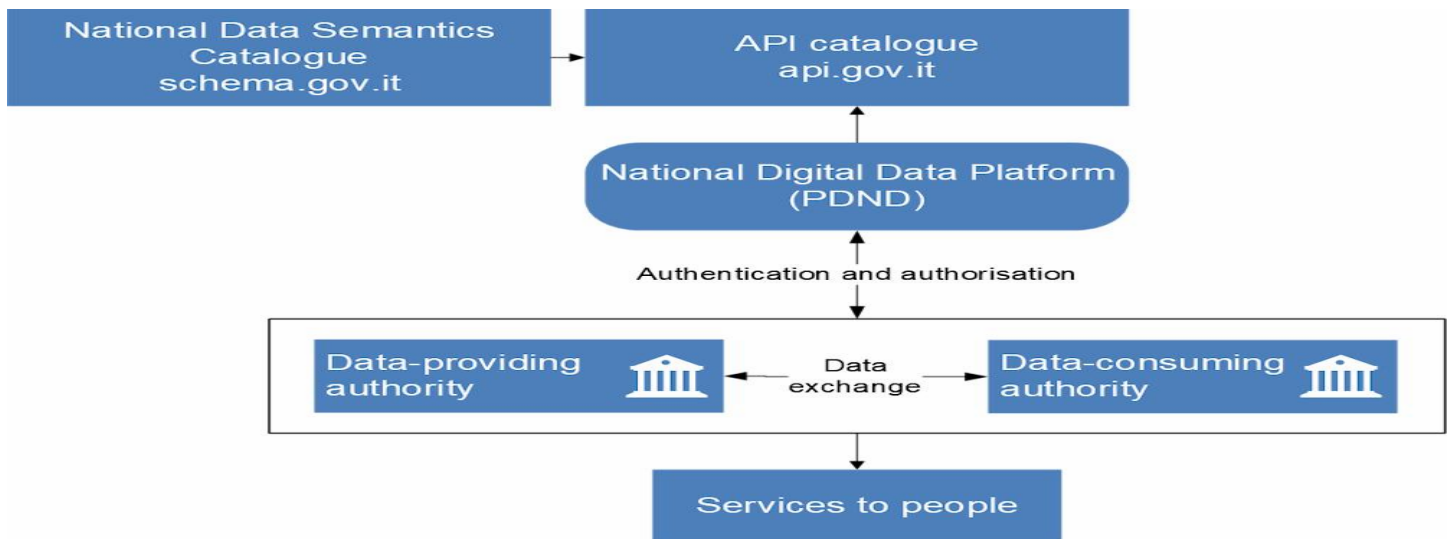
- Census data
- Economic data
- Geospatial information

Different departments must securely share data while maintaining privacy.

Objective

Design a governance and data management strategy.

Architecture Diagram



Governance Components

Component	Purpose
Data Lake	Central storage
Metadata Catalog	Data discovery
IAM	User authentication
Encryption	Data protection
API Gateway	Secure sharing

Data Management Strategy

Data Integration

- Apache NiFi
- ETL Pipelines

Metadata Management

- Apache Atlas

Data Quality

- Validation
- Deduplication
- Standardization

Master Data Management

- Unique citizen IDs
- Consistent records

Security Measures

Privacy

- Data masking
- Anonymization

Access Control

- Role-Based Access Control
- Multi-Factor Authentication

Encryption

- AES-256
- TLS

Interoperability

Government departments communicate through:

- REST APIs
- Standard data formats (JSON, XML)
- Shared metadata

Benefits

- Better decision-making
- Secure collaboration
- Improved public services

Task 4: Banking Fraud Detection Problem

Problem Statement

A bank processes millions of financial transactions every minute.

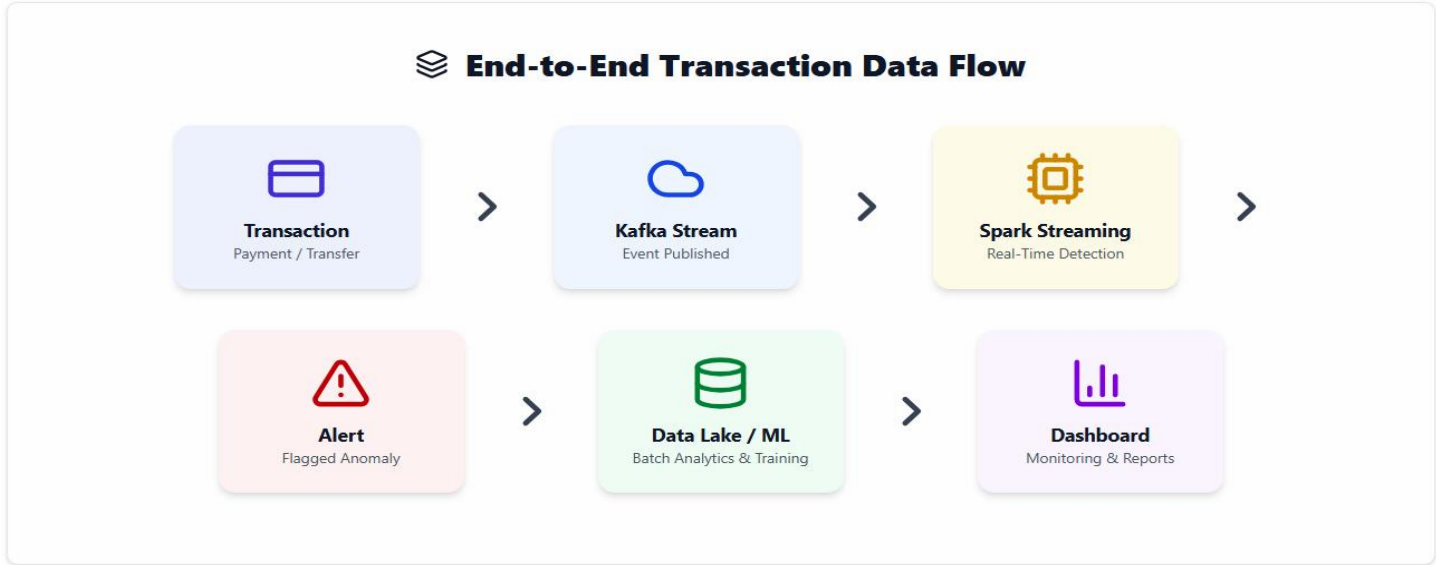
Requirements:

- Real-time fraud detection
- Low latency
- High accuracy

Objective

Design a scalable big data fraud detection pipeline.

Architecture Diagram



Architecture Components

Component	Technology
Transaction Stream	Kafka
Stream Processing	Spark Streaming
Historical Storage	HDFS
Database	Cassandra
ML Platform	Spark MLlib

Workflow

1. Transactions enter Kafka.
2. Spark analyzes transactions immediately.
3. Features are generated.
4. ML model predicts fraud.

5. Suspicious transactions trigger alerts.

Fraud Detection Algorithms

Classification

- Random Forest
- XGBoost
- Logistic Regression

Anomaly Detection

- Isolation Forest
- One-Class SVM

Deep Learning

- LSTM
- Autoencoders

Feature Examples

- Transaction amount
- Location
- Time
- Device ID
- Previous behavior

Benefits

- Prevents fraud losses
- Protects customers
- Reduces false positives

Task 5: Agriculture Smart Farming Problem

Problem Statement

Smart farms collect data from:

- Soil sensors
- Weather stations
- Drones

Farmers need recommendations for:

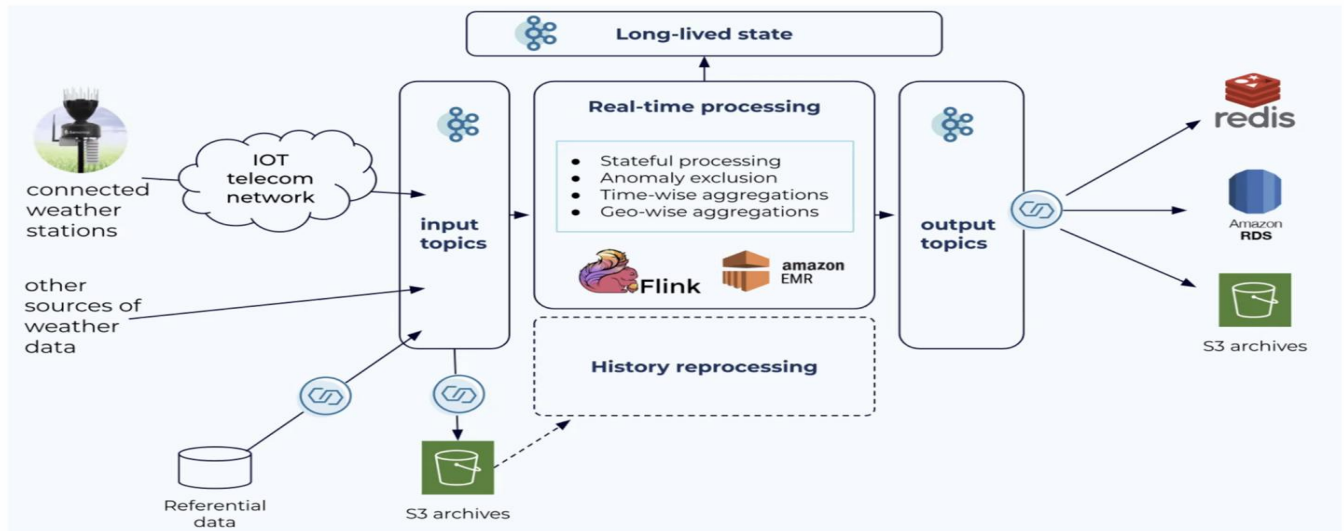
- Irrigation
- Fertilizer
- Crop health

Objective

Design a precision agriculture big data solution.

Architecture Diagram

Smart Agriculture with Kafka and Flink



Architecture Components

Component Technology

Sensors	IoT Devices
Drones	Camera Imaging
Ingestion	Kafka
Processing	Spark
Storage	Cloud Data Lake
Dashboard	Power BI

Workflow

1. Sensors collect soil moisture.
2. Weather stations send forecasts.
3. Drones capture crop images.
4. Kafka streams data.
5. Spark analyzes conditions.
6. Farmers receive recommendations.

Analytics Models

Yield Prediction

- Random Forest
- Gradient Boosting

Disease Detection

- CNN Image Classification

Irrigation Optimization

- Time Series Forecasting

Decision Support

Analytics provides:

- Water requirements
- Fertilizer recommendations
- Disease alerts
- Harvest timing

Benefits

- Higher crop yield
- Water conservation
- Reduced costs
- Sustainable farming

Comparison of All Five Solutions

Problem	Data Sources	Stream Tool	ML Models	Storage
Retail	POS, Online	Kafka + Spark	LSTM, K-Means	HDFS/S3
Social Media	Posts, Likes	Kafka + Flink	BERT, XGBoost	HDFS
Government	Census, GIS	NiFi	Data Quality Rules	Data Lake
Banking	Transactions	Kafka + Spark	Isolation Forest	Cassandra
Agriculture	IoT, Drones	Kafka	CNN, Random Forest	Cloud Storage

Common Big Data Technologies Used

Category	Technologies
Data Ingestion	Apache Kafka, Apache NiFi
Stream Processing	Apache Spark Streaming, Apache Flink
Batch Processing	Apache Spark
Storage	HDFS, Amazon S3, Cassandra
Machine Learning	TensorFlow, Spark MLlib, Scikit-learn
Visualization	Power BI, Grafana
Security	TLS, AES-256, RBAC

Conclusion

Each industry requires a scalable big data architecture capable of handling both real-time and historical data. By combining Apache Kafka for data ingestion, Spark/Flink for processing, cloud data lakes for storage, and machine learning models for analytics, organizations can achieve low-latency decision-making, high scalability, improved security, and more accurate predictions across retail, social media, government, banking, and agriculture.